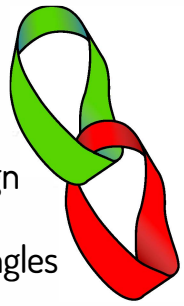
# MathsJam Shout

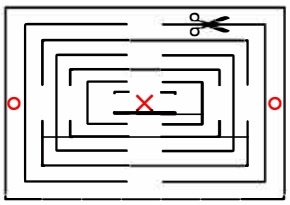
**December 2025**
**Various**

## Make
### Maths Decorations

Why not make your MathsJam pub temporarily more numerically festive with some mathematically-interesting Christmas decorations:

- Möbius paper chains
- Baubles made from printed polyhedra nets
- Variations on the design below, using different shapes instead of rectangles





Make several; attach the parts marked ○ to the previous one, and the part marked × to the next one

## Puzzle
### Lucky Numbers

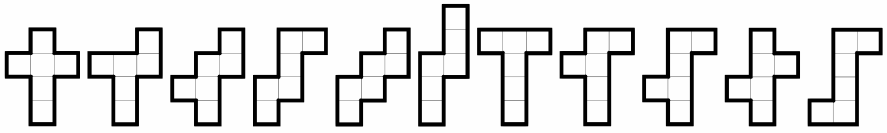
Credit: Vlad Mitlin, via Andrej Cherkaev

A 'lucky number' is a 6-digit number where the sum of the first three digits is the same as the sum of the last three digits.

Show that the sum of all possible lucky numbers is divisible by 13.

## Fun
### Come up with your own...

- Designs for the capital numbers (assuming the digits 0-9 we currently use are in lower-case)
- Charades or Articulate-style descriptions for maths words - use the generator at **tmip.uk/writingchallenge**
- Names for each of the eleven possible nets of a 3D cube:

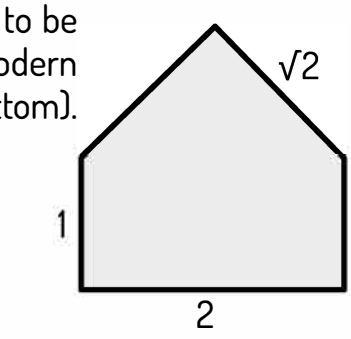


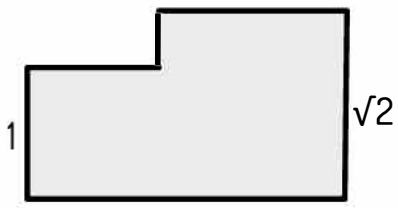
## Puzzle
### House Dissection

Credit: Robin Houston, based on original via Colin Wright, Thane Plambeck, David Klarner and N G de Bruijn; source unknown

An old-fashioned house (top) needs to be converted into a more efficient modern house, made from two squares (bottom). Can you find a way to do this by:

- Cutting it into **three** pieces and rearranging them?
- Cutting it into **three** pieces and rearranging them, in a different way that's not just a modification or symmetry of the first way?
- Cutting it into **two** pieces and rearranging them (harder!)




**MathsJam Shout** is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com** for details.

MathsJam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly MathsJam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest MathsJam, visit the website at **www.mathsjam.com**.